(Dis)patches from the WebPKI

Fina, Static CT, MTC, and PLANTS

Luke Valenta (Cloudflare), David Benjamin (Google), Matthew McPherrin (Let's Encrypt), Filippo Valsorda (Geomys)
RWC 2026

This talk

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And why you should care.

- Standardization efforts are underway at IETF
- Chrome to *only* support MTCs in new quantum-resistant root program
- Major CAs are interested
- Large-scale MTC feasibility study is ongoing
  - Early results (this talk)

Not just browsers.

- Ubuntu upki project opening a path for MTCs in Linux

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2024: 🩹 Static CT 🌞 (Filippo Valsorda, “Modern transparency logs”, RWC 2024)

- Log data served as static, cacheable assets
- Makes running CT logs cheaper and easier: already significant support in the CT ecosystem
- … but, upcoming changes will make running CT-based PKIs even harder

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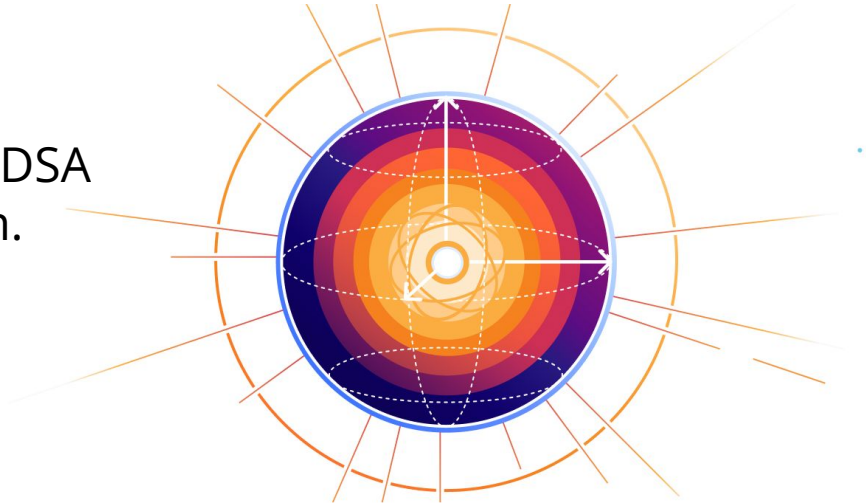
Static CT *helps*, but there are still fundamental scaling issues

- Cert explosion: Each issued cert typically results in 10+ entries across logs 💥
- CT log consumers need to download *all* entries from *all* logs 🤯

The quantum threat

Quantum computers that break RSA and ECDSA are coming, but we don’t know exactly when.



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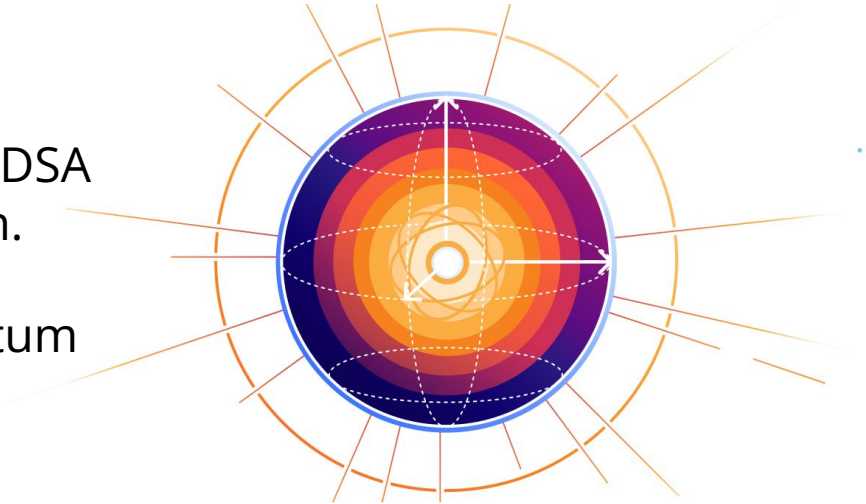
- For signing 🖊️: ML-DSA, SLH-DSA (+more in the pipeline)



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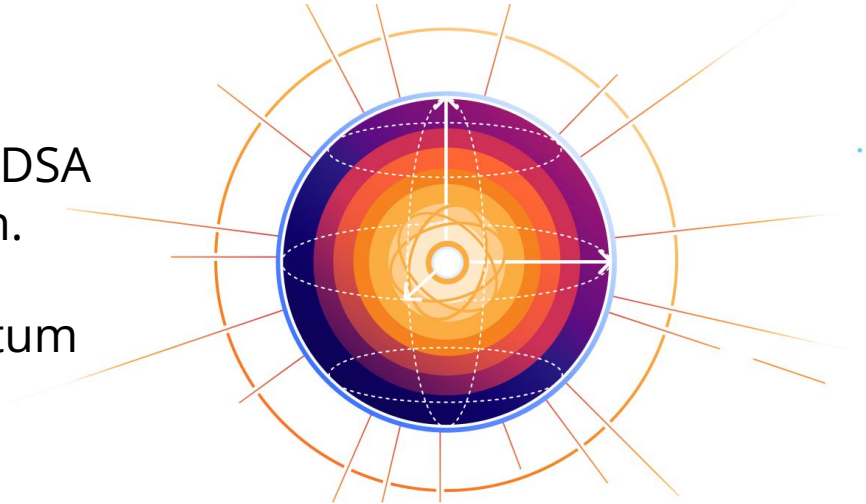

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The bad news: PQ signatures are big

- P-256 (ECDSA) keys are 64 bytes (uncompressed), and signatures 64 bytes
- ML-DSA-44 keys are 1,312 bytes (~20x), and signatures 2,420 bytes (~38x)

Post-quantum signatures are big

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  - With classical RSA and P-256, that's ~1.2kB
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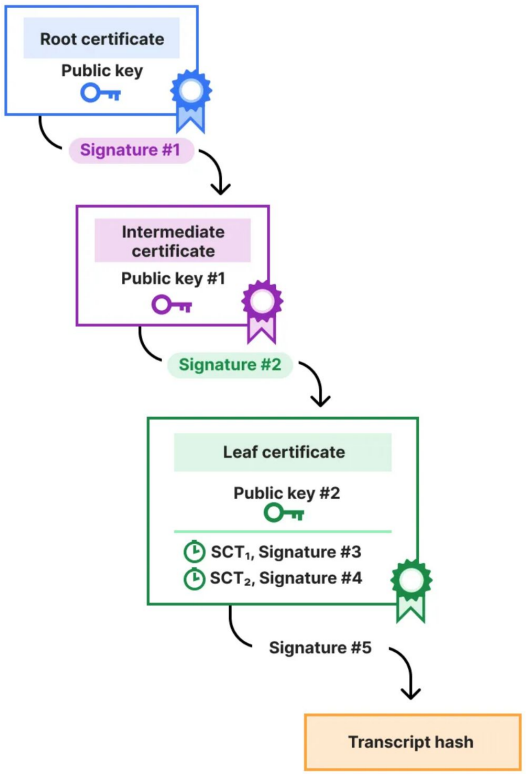

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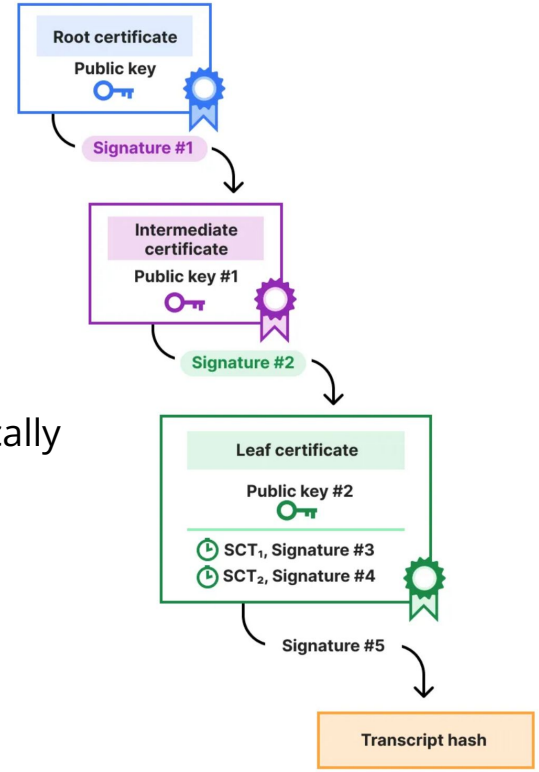

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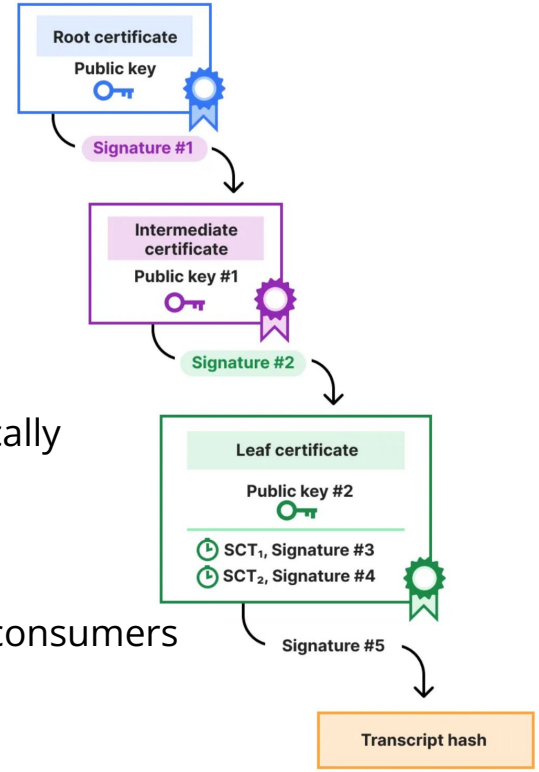

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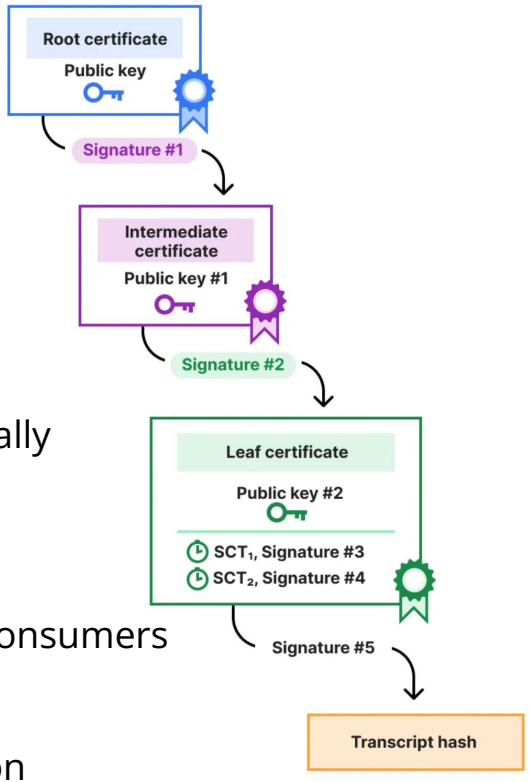

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- Do we need to drop CT support?
  - Chrome Security: CT helps to detect downgrades in PQ transition

WebPKI needs a *repotting, not more patches

Newly-formed PLANTS (PKI, Logs, And Tree Signatures) working group at the IETF is working on it. (*“Come for the PQ PKI, stay for the 🪴 puns.”)

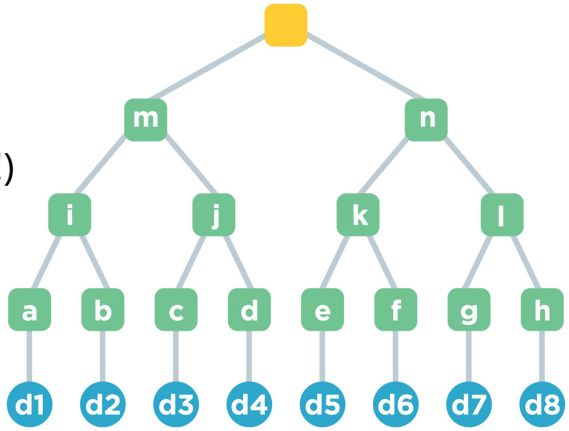
Goal: prune expensive PQ signatures from protocols like TLS, and build in transparency from the start.

First adopted work item is Merkle Tree Certificates (this talk).

Merkle Tree Certificates key ideas

“Don’t log what you issue, issue by logging”

- CAs run their own issuance logs (bonus: nice incentive alignment!)



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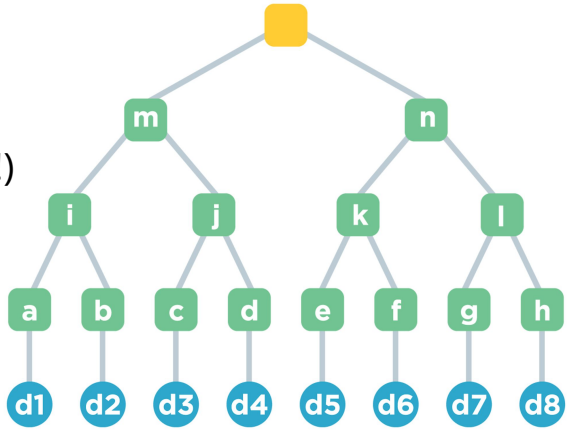

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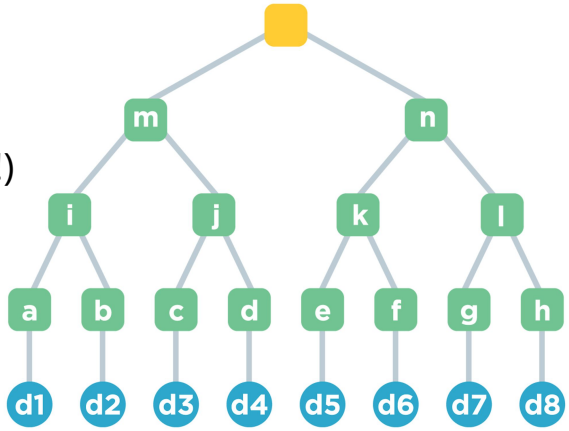

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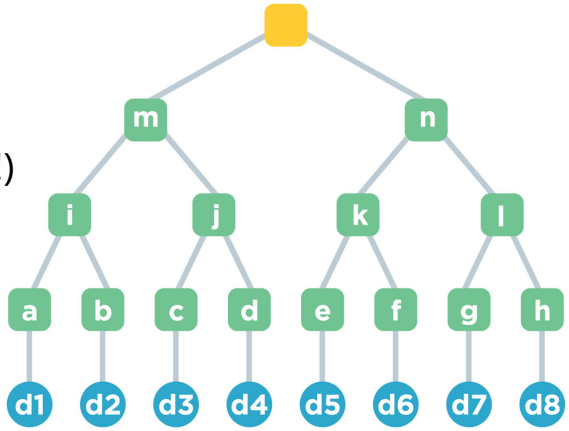

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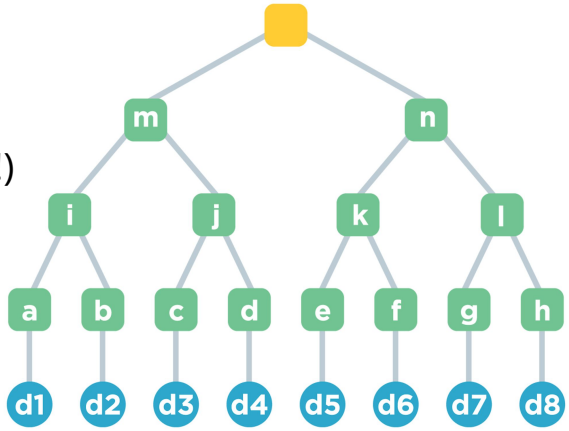

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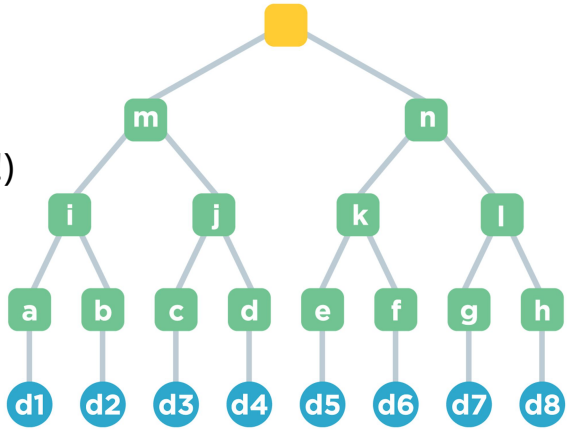

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(Learn more in David Benjamin’s in-depth technical overview)

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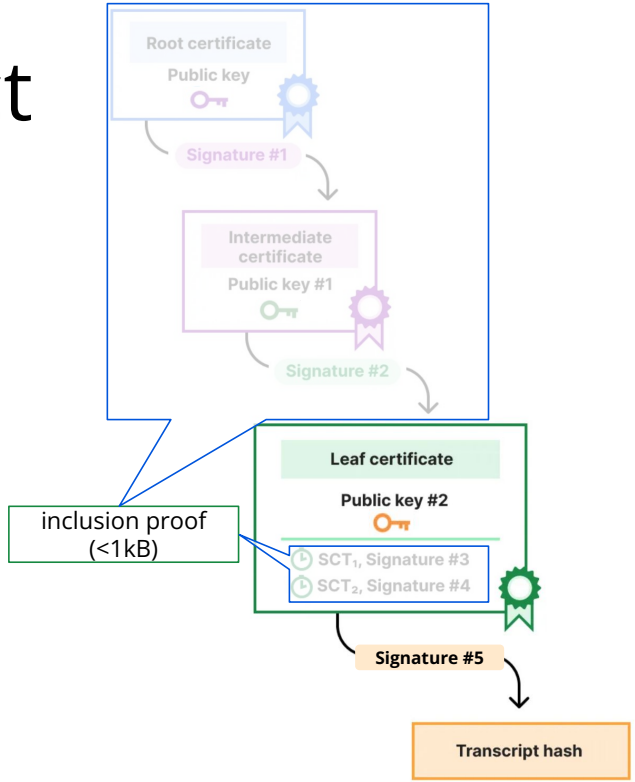

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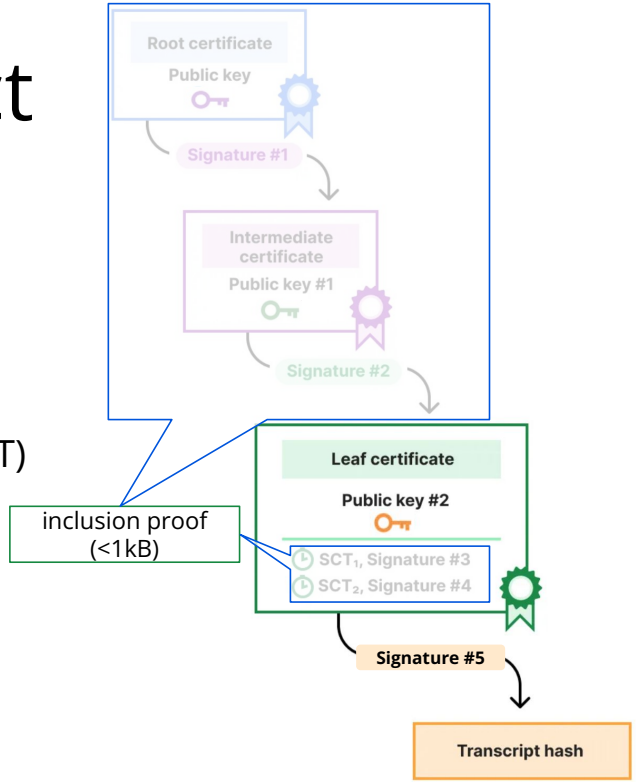

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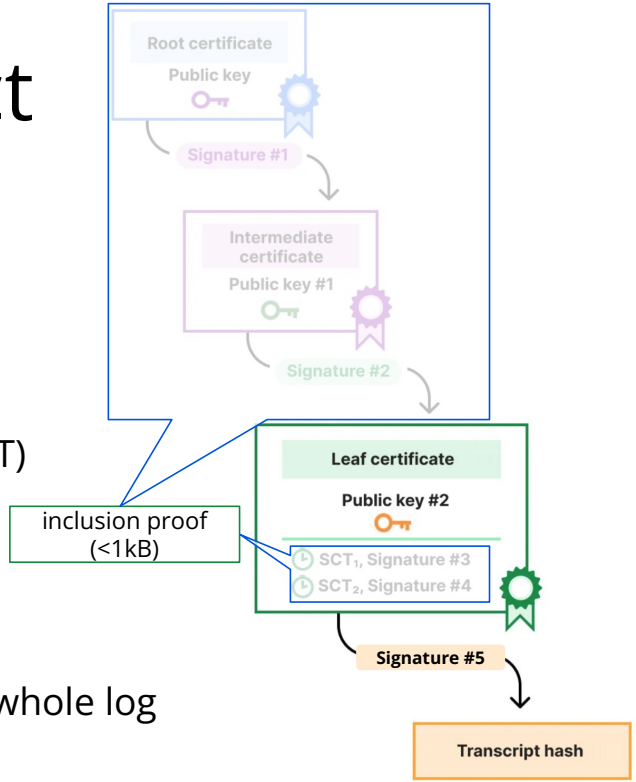

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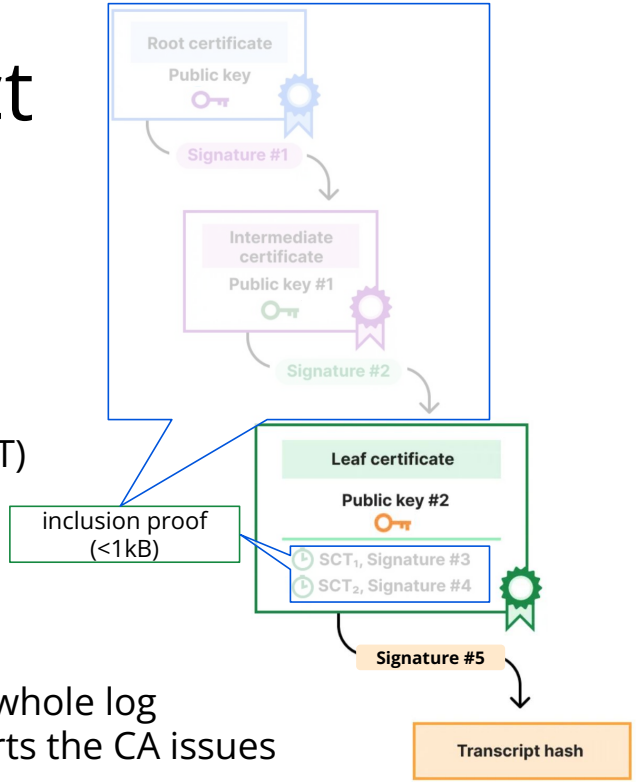

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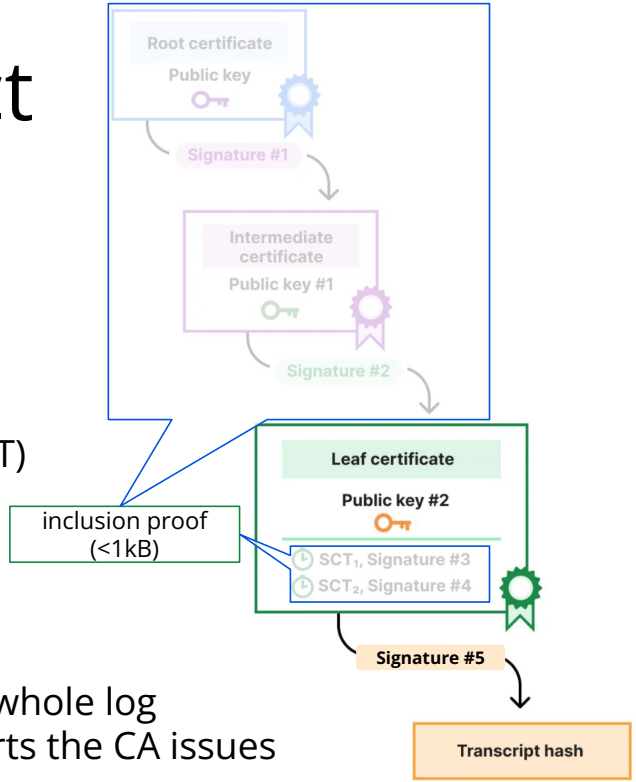

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- CA issuance log implementation is very similar to that of Static CT logs

Merkle Tree Certificates experiment

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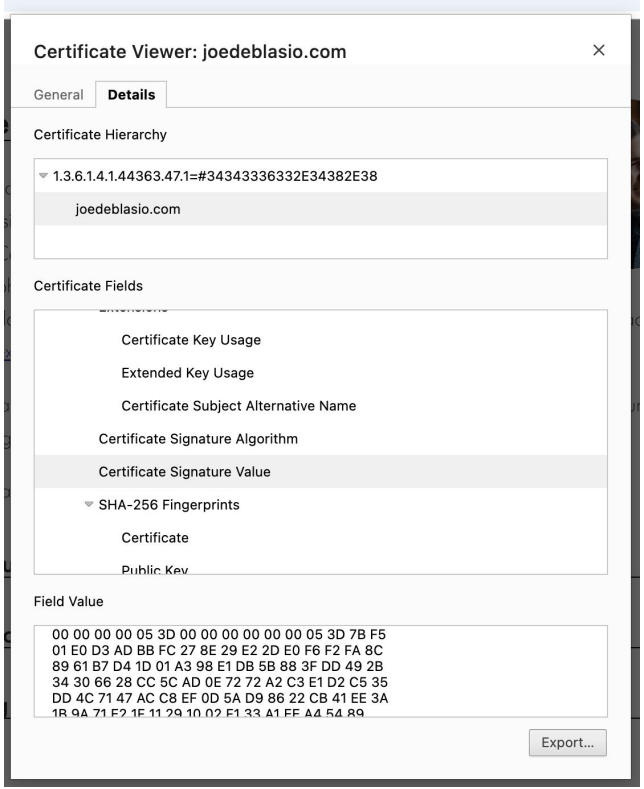
Cloudflare runs

- TLS server — currently 1000 Cloudflare-proxied domains
- "bootstrap" MTC CA
  - Every MTC is backed by a traditional cert chain
  - Restricts experiment to classical signatures
- Limit to landmark MTCs

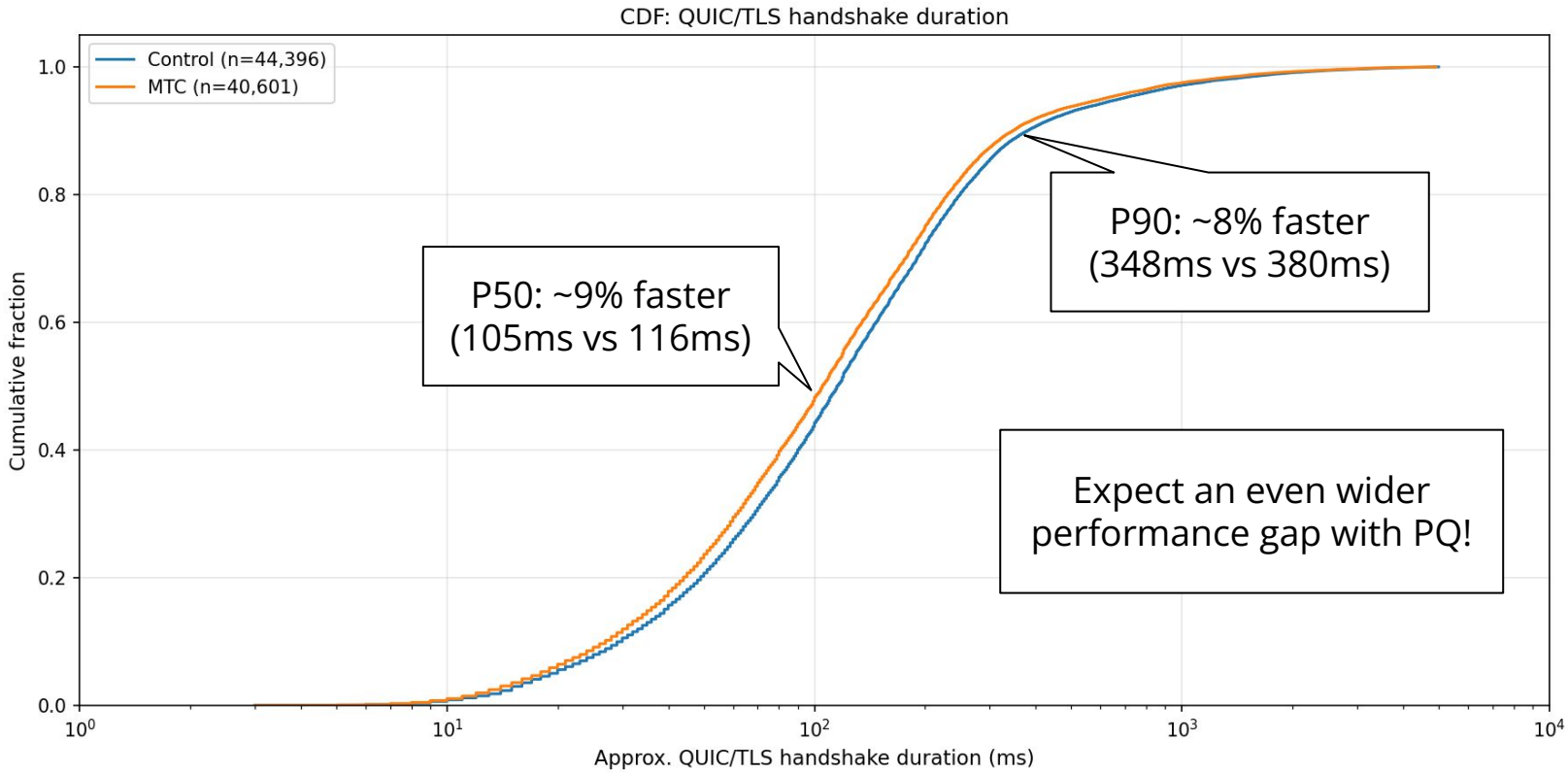
Chrome runs

- TLS client — currently 50% of Chrome Beta 146+
- Update service to validate and distribute landmarks

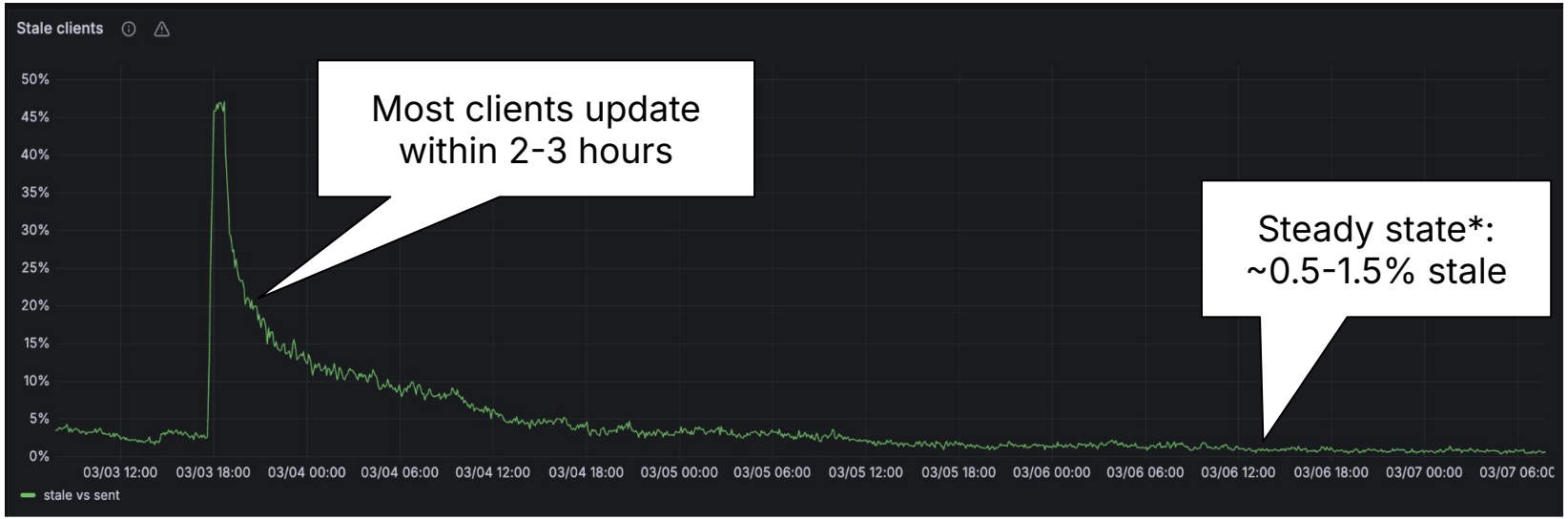


MTC in Chrome Beta

Are landmark MTCs faster than *classical*?



How quickly do clients get landmarks?



*This is an *upper bound* due to experimental quirks

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Operational experience

- We've had to shake out a few bugs on both client and server side
- Middlebox interference thus far is a non-factor (TLS 1.3 encrypts server cert)

What’s next?

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  - Implement and cross-test to help build interoperability and find corner cases
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